ECEN 5692 HW7 | Matt Hartnett

Problem 2.2.A.1

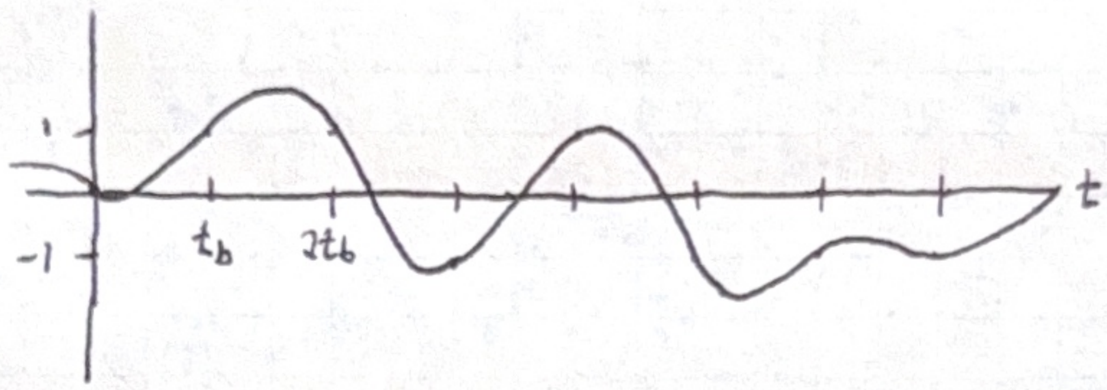
$$X(t) = \sum_{l=1}^{n} X_l\, u(t - l t_b)$$

1) $n=7$, X_l is BPSK, $u(t) = \mathrm{sinc}(t \frac{1}{t_b})$

Using a random number generator to find X_l:

$X_l = [1, 1, -1, 1, -1, -1, -1]$

$$X(t) = \mathrm{sinc}\left(\frac{t-t_b}{t_b}\right) + \mathrm{sinc}\left(\frac{t-2t_b}{t_b}\right) - \mathrm{sinc}\left(\frac{t-3t_b}{t_b}\right) \ldots$$
$$= \mathrm{sinc}(t/t_b - 1) + \mathrm{sinc}(t/t_b - 2) - \mathrm{sinc}(t/t_b - 3) \ldots$$

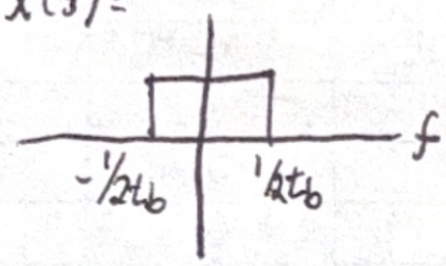


Note: I used a graphing calculator to find this plot

For linear stream modulation:

$$\tilde{X}(f) = \sum_{l=1}^{n} X_l e^{-j2\pi f l t_b}\, \tilde{u}(f)$$
$$\tilde{U}(f) = \sqrt{t_b}\, \mathrm{rect}(t_b f)$$

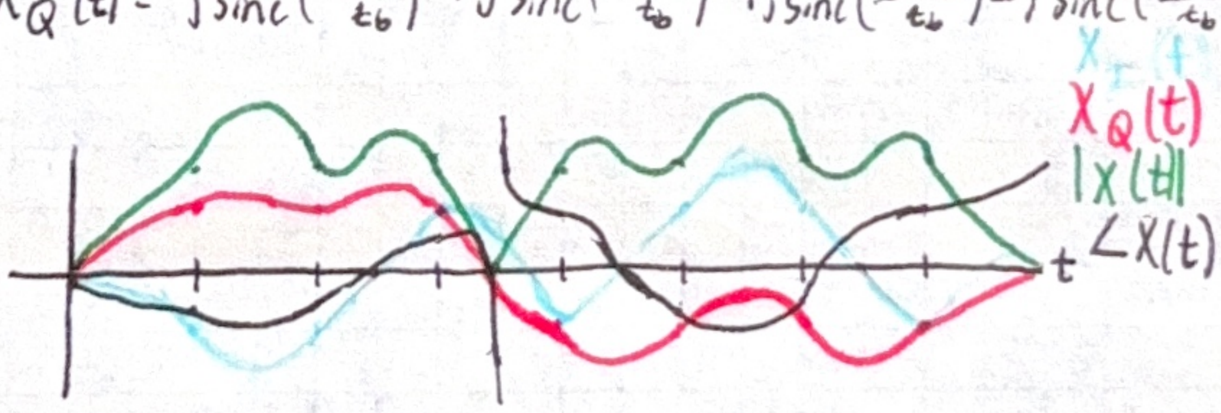
$\tilde{X}(f) =$



2) $n=7$, X_l is QPSK, $u(t) = \mathrm{sinc}(t\, 1/t_b)$

$X_l = [-1+j, -1+j, 1+j, -1-j, 1-j, 1-j, -1-j]$

$$X_I(t) = -\mathrm{sinc}\left(\frac{t-t_b}{t_b}\right) - \mathrm{sinc}\left(\frac{t-2t_b}{t_b}\right) + \mathrm{sinc}\left(\frac{t-3t_b}{t_b}\right) + \ldots$$
$$X_Q(t) = j\,\mathrm{sinc}\left(\frac{t-t_b}{t_b}\right) + j\,\mathrm{sinc}\left(\frac{t-2t_b}{t_b}\right) + j\,\mathrm{sinc}\left(\frac{t-3t_b}{t_b}\right) - j\,\mathrm{sinc}\left(\frac{t-4t_b}{t_b}\right) + \ldots$$



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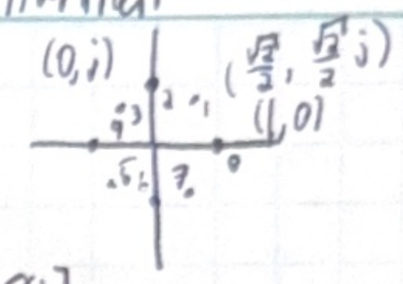
Problem 2.2.A.1 (cont)



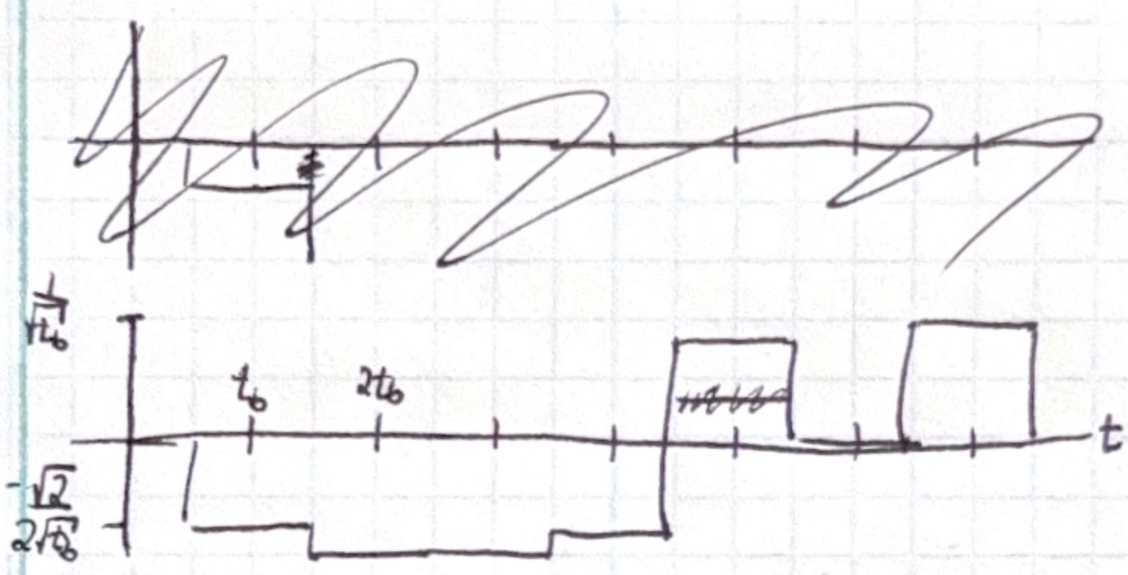
2) $n=7$, X_k is 8-PSK, $\mathcal{N}(t) = \frac{1}{\sqrt{t_b}} \text{rect}(t/t_b)$

$X_k = [-\sqrt{2}/2 - \sqrt{2}/2\, j,\ -1+0j,\ -1+0j,\ -\sqrt{2}/2 + \sqrt{2}/2\, j,\ \sqrt{2}/2 + \sqrt{2}/2\, j,\ 0+j,\ 1+0j]$

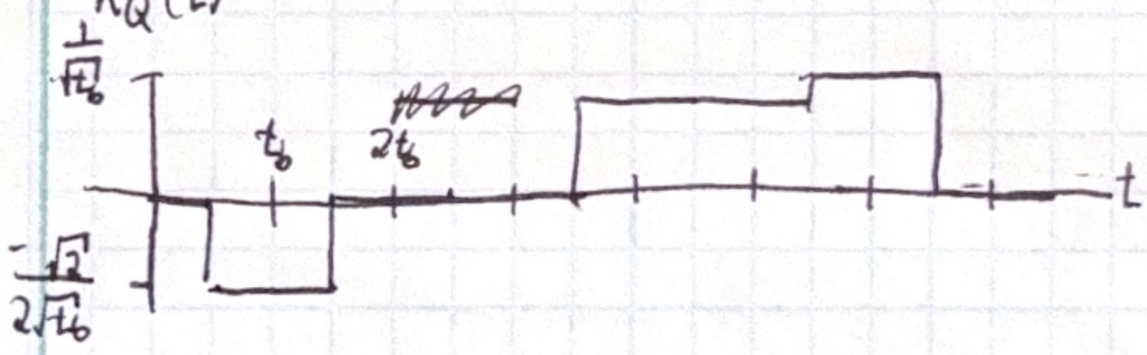
$X_I(t) = 1/\sqrt{t_b}\left(-\sqrt{2}/2\, \text{rect}((t-t_b)/t_b) - \text{rect}\left(\frac{t-2t_b}{t_b}\right) - \text{rect}\left(\frac{t-3t_b}{t_b}\right) - \frac{\sqrt{2}}{2}\text{rect}\left(\frac{t-4t_b}{t_b}\right) + \frac{\sqrt{2}}{2}\text{rect}\left(\frac{t-5t_b}{t_b}\right) + \text{rect}\left(\frac{t-7t_b}{t_b}\right)\right)$

$X_Q(t) = 1/\sqrt{t_b}\; j\left(-\sqrt{2}/2\, \text{rect}\left(\frac{t-t_b}{t_b}\right) + \frac{\sqrt{2}}{2}\text{rect}\left(\frac{t-4t_b}{t_b}\right) + \frac{\sqrt{2}}{2}\text{rect}\left(\frac{t-5t_b}{t_b}\right) + \text{rect}\left(\frac{t-6t_b}{t_b}\right)\right)$

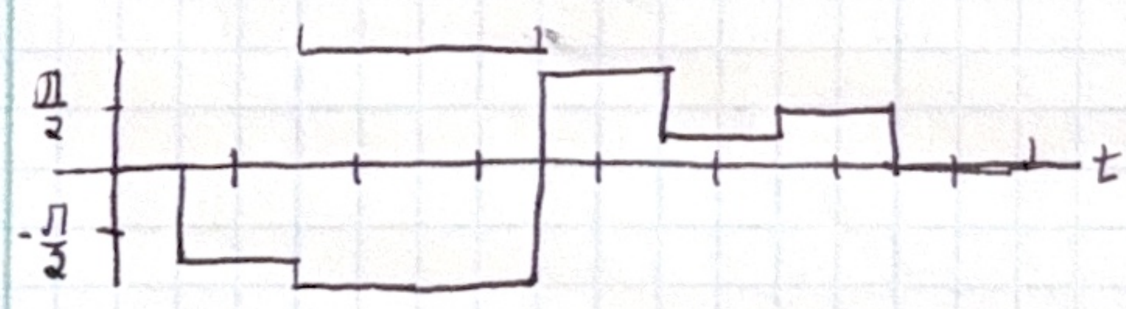
$X_I(t)$:



$X_Q(t)$:



$\angle x(t)$:



$|x(t)| = 1/\sqrt{t_b}$ for all values of t